

GROWIFY : *Preliminary Plan*

This project aims to optimize ad spend across multiple channels through **media mix modeling**. The **goal** is to establish a model that calculates the **optimal budget allocation** for brands by analyzing return on investment (ROI) and spend-data from the previous week. The model identifies the **best-performing channels** and recommends increasing **spend accordingly**. The project also includes analyzing the pull forward effect of promotions on sales, conducting market basket analysis to understand **customer purchase journeys**, and developing a **revenue forecasting model**. The project utilizes various metrics, such as total sales and the saturation curve, to provide insights into ad spend effectiveness and guide budget allocation decisions. The output of the project includes **actual and predicted sales** with a 95% confidence interval, aiding in planning and decision-making. The project combines **media mix modeling, pull forward effect analysis, market basket analysis, and revenue forecasting** to **optimize ad spend** and make data-driven decisions.

PROJECT PLAN

1. Project Initiation:

- Define project objectives and scope.
- Identify key stakeholders and establish communication channels.
- Formulate a project team with relevant expertise - roles and tasks
- Set project timelines and milestones.
- Define project success metrics based on per model.
- Design structure for A/B testing for evaluating models.

Time required: 1 week

2. Data Collection and Analysis:

- Gather historical data on ad spend, sales, and other relevant metrics. (2-3 days)
- Clean and preprocess the data for analysis.
- Conduct exploratory data analysis to gain insights into trends and patterns.
- Identify any data gaps or issues that need to be addressed.
- Optimizing Spend, Saturation Curve by Channel, Trend Component, Seasonal Components and so on.

Time required: 2.5 weeks

3. Media Mix Model Development:

- Develop a media mix model to optimize ad spend allocation.
- Implement statistical and analytical techniques to calculate ROI and determine the best-performing channels.
- Validate the model by comparing predicted outcomes with actual results.
- Fine-tune the model based on feedback and iterative analysis.

Time required: 3 weeks

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4. Pull Forward Effect Analysis:

- Build a model to analyze the impact of promotions on sales.
- Identify sales spikes, dips, and the hangover period.
- Calculate time distortion and assess the positive or negative impact on sales and marketing channel effectiveness.
- Incorporate the pull forward effect analysis into the media mix model.

Time required: 2.5 weeks

5. Market Basket Analysis:

- Perform market basket analysis to understand customer purchase journeys.
- Identify product combinations that have the highest impact on sales and revenue.
- Analyze the effectiveness of different campaigns within channels.
- Integrate market basket analysis results into the media mix model.

Time required: 3 weeks

6. Revenue Forecasting:

- Develop a revenue forecasting model based on historical data and relevant factors.
- Incorporate sales trends, seasonality, spend levels, and holidays into the model.
- Validate the model using historical data and assess its accuracy.
- Generate forward-looking revenue forecasts with confidence intervals.

Time required: 3 weeks

7. Project Evaluation:

- Evaluate the performance and effectiveness of the developed models.
- Assess the impact of optimized ad spend allocation on revenue and ROI.
- Analyze the accuracy and reliability of the revenue forecasting model.
- Identify any limitations or areas for improvement.
- Design A/B Testing experiments or multivariate experiments to identify model success and growth

Time required: 2 week - overall (depends on how long A/B tests need to be run)

8. Documentation and Reporting:

- Prepare a comprehensive project report documenting the entire project lifecycle, including objectives, methodologies, findings, and recommendations.
- Include visualizations, charts, and tables to present the results clearly.
- Provide an overview of the developed models and their implementation.
- Discuss the impact of the project on optimizing ad spend and improving revenue outcomes.
- Highlight any challenges faced during the project and propose future research or enhancements.
- Design future experiments involving advanced models for recommendation engines

Time required: 1 week